Algorithmic Collusion Without Deterrence: Auditing the Punishment Mechanism in Q-Learning Pricing Agents

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Abstract

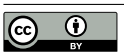
Independent tabular Q-learning agents repricing in a simulated Bertrand duopoly converge to supracompetitive prices, and the seminal evidence attributes this to a learned reward–punishment scheme: undercut your rival and it retaliates, so deviation never pays. This mechanistic reading is what turns high prices into a cartel, and into a regulatory target. We replicate the environment of Calvano et al. [2020] and confirm their impulse-response pattern within co-trained pairs. We then audit the pattern on frozen policies, with learning off, and find that on its own it does not identify a mechanism: the return is over-determined rather than selective. Seventeen of eighteen converged pairs have a single attractor whose basin is the entire state space, so every perturbation returns whatever provoked it, and a forced price *increase* triggers the same retaliatory pattern as an undercut. What co-training produces is a narrow, dyad-specific basin of attraction: re-match frozen agents from independent runs, as Calvano et al. did with learning on, and coordination collapses by 81% with nothing carried across. The deterrence claim is incomplete on its own terms: the one-period undercut of Calvano et al.'s impulse-response test is unprofitable, which we confirm in every pair, but the *optimal* deviation, computed by dynamic programming against each partner's frozen strategy, profits by 14.8% on average. The scheme deters only the deviation no optimizer would choose; collusion survives because deviation is never tried, not because it would not pay. Supracompetitive prices are robust; the punishment mechanism read into them does not survive these tests, which changes what the standard test of algorithmic collusion establishes, and what pairing-disruption remedies can be expected to do.

CCS Concepts

• **Computing methodologies** → **Multi-agent reinforcement learning**; • **Theory of computation** → *Solution concepts in game theory*; • **Applied computing** → *Economics*; Law.

Keywords

algorithmic collusion, multi-agent reinforcement learning, Q-learning, algorithmic pricing, market simulation, antitrust



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1 Introduction

A growing literature reports that simple reinforcement-learning agents, trained independently in repeated pricing games, converge to prices well above the static Nash equilibrium. The seminal contribution of Calvano et al. [2020] goes further than documenting supracompetitive prices: it offers a mechanism. After a forced unilateral price cut, the rival's price drops and both prices then return to their pre-deviation level, a pattern the authors read as a "reward–punishment scheme" that stabilizes the collusive outcome. This mechanistic reading is what gives the result its policy weight: prices that are merely high are an artifact; prices *defended by retaliation* are a cartel.

This paper asks a narrow question with broad consequences: **is there actually a punishment mechanism?** We take the question seriously on Calvano et al.'s own terms. We replicate their environment and their impulse-response experiment, and we confirm their pattern: within co-trained pairs, the deviation-drop-return dynamic is robust (Section 4.2). The contribution builds on a line of work that has questioned parts of the mechanistic reading with separate instruments (Section 2), and it is twofold. The negative half is a five-part audit of the mechanism behind the pattern (Section 4.3) and an incentive accounting of the converged play (Section 4.5), run jointly on the same frozen policies: taken on its own terms, the exhibit does not establish deterrence. The positive half is what we find in its place: co-training produces a *dyad-specific basin of attraction*, a joint object carved by the pair and without value outside it. The direct test takes the rematching exercise of Calvano et al. [2020, Section VI.B] and turns it into an evaluation protocol: re-match converged agents across independent training runs with learning off, and coordination collapses by 81% (Section 4.4). Throughout, the design separates the *learning phase* from the *measurement phase*: in the language of experimental economics, Calvano et al. evaluate in the partner treatment, and we add the stranger treatment as a test rather than as a re-training exercise (Section 3.1).

None of the five audits finds a punishment mechanism. The retaliatory response fires for *any* off-path price, including a cooperative price increase; faced with an opponent that actually punishes, the agents undercut it and absorb the permanent punishment rather than sustain the cooperative path it offers; the learned reaction functions are uncorrelated with best-response logic; the celebrated

return-to-baseline is forced by the geometry of the converged policies, which have a single global attractor, so it would follow any perturbation; and none of this reflects a representational limit, because hand-built punishment rules fit the same policy matrix exactly. What survives the audit is real but different in kind: a locally calibrated, dyad-specific basin of attraction, fast to return, blind to why it was perturbed, and specific to the pair that carved it. Deterrence is a property an agent has; what training built here is a property only the pair has.

We are careful about what we do *not* claim. Calvano et al. report that about half of their sessions reach an on-path Nash equilibrium of the repeated game, with small one-step losses elsewhere, and rarely a subgame-perfect one; we do not dispute these statistics, and Section 4.5 says how our best-response measure relates to theirs. Our target is the mechanistic interpretation, the "reward–punishment scheme," in the sense that matters for economics: a scheme, in the tradition of Friedman [1971] and Abreu [1988], is a rule that *enforces a prescribed path*. It responds to departures from that path, and it sustains the path because the response makes deviating unprofitable. That enforcement structure is what the audits do not find.

2 Background

Punishment schemes. The language of "reward–punishment schemes" carries a precise meaning in repeated-game theory. In Friedman [1971], collusion is sustained by a trigger strategy: cooperate until anyone defects, then revert to static Nash forever. Abreu [1988] generalizes this to optimal penal codes, in which deviations, including deviations from the punishment itself, restart a calibrated punishment phase. Two features are constitutive. First, *path enforcement*: the scheme prescribes a path and responds to departures from it, in order to make deviating unprofitable. Second, *incentive compatibility*: the threat is credible because carrying it out is optimal given the scheme (under imperfect monitoring, punishment can even fire without any deviation; Green and Porter [1984]). Our audit therefore does not ask whether the learned response fires only after defections (a trigger rule need not be selective in that sense) but whether the response has the *structure* of path enforcement (Sections 4.3(i)–(iii)), whether the enforced path is incentive-compatible (Section 4.5), and whether the return itself is selective at all, or forced by the structure of the state space (Section 4.3(iv)).

The Calvano et al. [2020] result. Two firms in a logit-Bertrand duopoly simultaneously choose prices from a 15-point grid; demand is $q_i = e^{(a_i - p_i)/\mu} / \left(\sum_j e^{(a_j - p_j)/\mu} + e^{a_0/\mu} \right)$ and profit is $\pi_i = (p_i - c)\, q_i$, with $a = 2$, $c = 1$, $\mu = 0.25$, $a_0 = 0$. Each agent runs tabular Q-learning [Watkins and Dayan 1992] on the state $s_t = (p^A_{t-1}, p^B_{t-1})$ (225 states) with $\alpha = 0.15$, $\delta = 0.95$, and ε-greedy exploration decaying geometrically to zero. After convergence, prices settle well above Nash. We measure this with the collusion index $\mathrm{CI} = (\bar{p}^{RL} - p^{NE})/(p^M - p^{NE})$, where $p^{NE} \approx 1.473$ and $p^M \approx 1.925$. Calvano et al.'s impulse-response analysis (force one agent to undercut for one period; the rival's price falls and both then recover) is the exhibit read as a "reward–punishment scheme," and it is what elevated the result to a landmark for competition policy.

Subsequent literature and the gap. Klein [2021] finds supracompetitive outcomes under sequential pricing, sensitive to action-space granularity. Asker et al. [2024] show that, for myopic agents, counterfactual ("synchronous") updating drives prices to the competitive level, so that high prices stem from what the learning protocol fails to evaluate rather than from strategy; Hansen et al. [2021] obtain supracompetitive prices from bandit algorithms with *no* punishment capability, and Abada and Lambin [2023] reach a similar exploration-based diagnosis. Empirically, algorithmic pricing changes price competition [Assad et al. 2024; Brown and MacKay 2023]; the regulatory stakes are drawn out by Harrington [2018] and OECD [2017]. Closest to us, Calvano et al. [2023, hereafter Calvano IJIO] verify approximate incentive compatibility on converged profiles and classify the 2020 environment's collusion as largely "genuine." The interpretation has since been examined from several directions, each with its own instrument. Epivent and Lambin [2024] show, in the same environment, that the price war follows price *hikes* as well as cuts, so the response does not distinguish the direction of the deviation. den Boer et al. [2026] show that the impulse-response pattern (the rival cuts, prices return, the deviation does not pay) can be generated by strategy profiles that are neither equilibria nor reward–punishment schemes, and conclude that a systematic test would have to check every unilateral deviation; they carry this out exhaustively for two prices, and the 15-price case is beyond enumeration. Banchio and Mantegazza [2023] describe a mechanism that sustains supracompetitive prices without any punishment: correlated estimation errors across co-learning agents, which they call spontaneous coupling. And Calvano et al. [2020, Section VI.B] themselves rematch converged agents with learning still on: profit gains fall from 85% to about 20% before re-learning partially restores them, which they read as coordination being "almost completely pair-specific." Tang [2024] surveys this debate under the labels *spurious*, *genuine* and *authentic* collusion, the last being Tang's name for the definition of den Boer [2023].

What has been missing is a joint audit, on the same converged policies and with learning off, that applies these instruments together and adds the three the debate still lacks: an explicit random-policy null for the return itself, a capacity control for whether the architecture could express a punishment scheme at all, and an exact best response against each converged partner, the equilibrium check den Boer et al. perform by enumeration for two prices, carried out in the 15-price case. The cross-pairing control turns Calvano et al.'s rematching exercise into an evaluation protocol, and connects it to the brittleness documented in multi-agent RL as joint policy correlation, the overfitting of independently trained policies to their co-learners [Lanctot et al. 2017].

3 Experimental Design

3.1 Training and testing: the logic of the design

Every result in this paper rests on the separation between a *learning phase* and a *measurement phase*. The training stage (learning ON) is not the object of study; it is the process that generates the objects of study, the analogue of the early rounds of a laboratory experiment, or of the estimation sample in econometrics. At convergence, the Q-tables are frozen and play becomes greedy: the test stage (learning

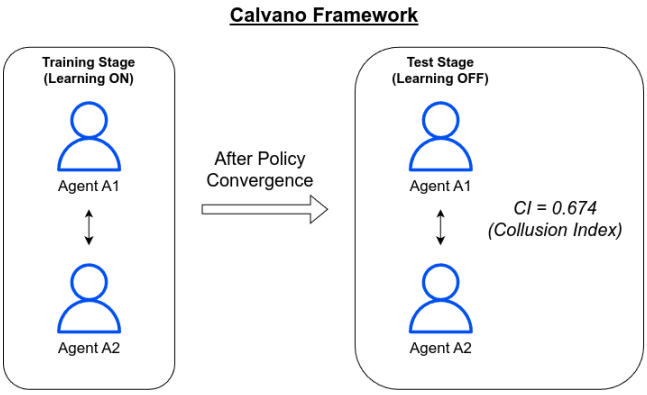


Figure 1: The standard evaluation framework of the algorithmic collusion literature [Calvano et al. 2020]: the pair of agents that learns together (training stage, learning ON) is the same pair that is evaluated after policy convergence (test stage, learning OFF): the partner design. All impulse-response evidence is collected within the original training dyad; training and testing contexts are never separated. The collusion index shown (CI = 0.674) is the within-pair value from our replication.

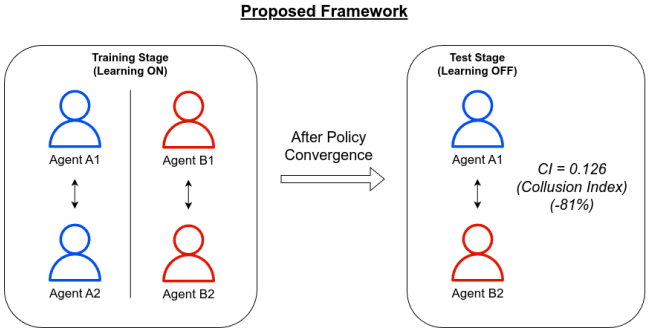


Figure 2: The stranger test: Calvano et al.'s rematching confined to the test stage, with learning off. Training is identical to Calvano et al.'s, replicated in independent runs (run A, run B); the manipulation is confined to the test stage: pairing an agent from run A with an agent from run B it never interacted with. If co-training produced individual collusive strategies, coordination would survive the re-pairing; if it produces a dyad-specific joint basin, it collapses. It collapses: CI falls from 0.674 (within-pair, Figure 1) to 0.126 (−81%; Section 4.4).

OFF), the analogue of out-of-sample validation. Every number in this paper is collected in the test stage.

This separation is where the literature's evidence is thinnest. In the standard framework (Figure 1), the pair that learns together is the same pair that is evaluated after convergence: all impulse-response evidence, in Calvano et al. [2020] and in the studies that followed it, is collected *within the original training dyad*, the *partner design* of experimental economics [Andreoni 1988]. Nothing in that design can distinguish an individual strategy, which an agent carries with it, from a joint convention, which exists only inside the pair. In machine-learning terms, the standard test evaluates each policy exactly on the distribution of states its own training partner generates: an in-sample measurement, presented as evidence about the policy itself. But a mechanism claim is a claim about the *agent*: that it carries a rule that detects and punishes deviations. Testing that claim requires moving the agent out of the context that produced it. Calvano et al. [2020, Section VI.B] do move it, as a re-training exercise with learning on; what had not been done is to use the move as a test, with learning off, so that what the policy carries is separated from what re-learning recovers.

Our addition is to make that manipulation a test: a one-shot *stranger* evaluation of frozen converged play (Figure 2). After convergence, an agent from run A is paired with an agent from run B, a partner it never interacted with. The treatment is the pairing, not the training, and the question it answers is the classic rematching question: is the observed behavior an individual strategy, or a convention of the pair? Freezing is not a simplification but the regime that matters for deployment: the policy that reaches a market is the one trained offline, and Calvano et al.'s rematched agents re-coordinate only after tens of thousands of further periods of greedy play at live prices against the new competitor (their Figure 11; fast relative to the original training, as they note, but paid in realized profit). The frozen test asks what the policy carries before that bill is paid.

3.2 Environment, training, and protocols

Environment and training. Bertrand-logit duopoly as in Section 2, 15-point grid spanning the Nash and monopoly benchmarks. Agents are trained in pairs ("co-trained") for 2M steps. Each run produces one co-trained pair. The core population is 18 pairs; a post-hoc seed audit found one run bit-identical to another (a resume-dependent seed formula in the runner, since fixed), so statistics use the 17 unique pairs; after the fix, retraining and six further runs extended the population to 24 unique pairs, and results that use the extended population say so explicitly. The exact-DP audit of Section 4.5 adds a high-fidelity 10M-step reference pair (38 agents in total). All post-training evaluations are greedy (no learning, no exploration). Deliberate divergences from Calvano et al.'s protocol (a fixed 2M-step budget with 10,000-step episode resets and random initial states, versus their infinite horizon and 100,000-period stability criterion) are controlled for by a separate five-pair population trained under the faithful regime (no resets, single horizon, stability stopping rule), on which Section 5 reruns the entire audit chain. In total the paper draws on 24 co-trained pairs at 2M steps, one 10M-step reference pair, five pairs under the faithful regime, 20 seeds per grid size in the granularity sweep of Section 4.6, and 1,024 pairs on Calvano's grid for the cross-pairing check at scale in Section 4.4; every audit is computed exactly on the converged policies, pair by pair, rather than averaged over sessions.

Reproducibility. The environment, agents, training and audit scripts, configurations, seeds and the converged Q-tables behind every figure are available at https://github.com/kauelopes/algorithmic-collusion-without-deterrence.

Protocols. (i) *Impulse response*: after the pair settles, one agent's price is exogenously forced for one period and both prices are recorded over a [−5, +40] window; shocks go *down* (to the static best response, Calvano et al.'s deviation) and *up* (to the monopoly price); each cell runs under two arms, co-trained and a *cross-paired* control, with 10 random initializations per pair. (ii) *Fixed-opponent evaluation*: each trained agent plays nine scripted strategies; a dedicated opponent-model test initializes the joint state at the cooperative price, so that trigger strategies start untriggered, and plays each agent 200 periods against grim-trigger, constant-Nash and constant-monopoly opponents. (iii) *Cross-deployment*: agents from different runs are matched and evaluated greedily. (iv) *Exact dynamic programming*: fixing the partner's greedy policy induces a 225-state MDP; we compute the exact value of converged play and of the optimal best response; the partner *reacts*, so this is not a one-shot counterfactual. (v) *Null model*: the impulse-response protocol applied to pairs of random Q-tables drawn on the same value range. (vi) *Capacity control*: hand-constructed punishment policies embedded in the same 225 × 15 matrix.

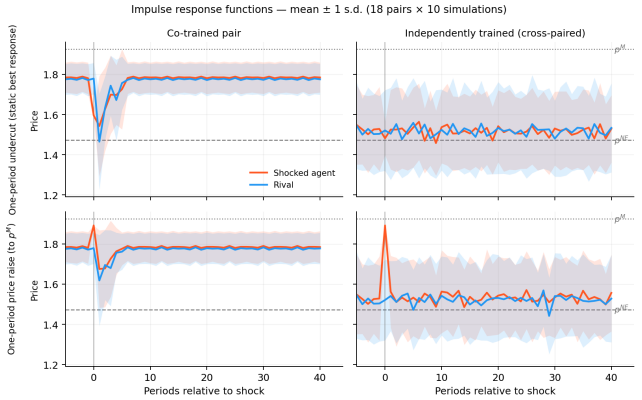


Figure 3: Impulse-response functions, mean ± 1 s.d. Rows: downward shock (static best response) and upward shock (to pˆM); columns: co-trained pair vs cross-paired control. Within pairs (left), both shocks produce the drop-and-return pattern; across pairs (right), the shock is invisible against the ambient variation.

4 Results

4.1 The phenomenon replicates

Everything we go on to audit, we first reproduce. Across 50 seeds at 2M steps, the mean collusion index is **CI = 0.6245** (CI = 0 at the Nash price, 1 at the monopoly price), within the range spanned by published replications; a single high-fidelity run at 10M steps yields CI = 0.6421 (the two use different grid widths, 1.0 for the 50-seed batch and Calvano's 0.1 for the reference run, so the two indices are not directly comparable). Our environment reproduces the phenomenon we set out to examine. One number recorded here becomes important later: in every run of this project in which it is measured on the converged attractor (147 runs), the counterfactual deviation gain (CE-gap) is strictly positive (0.014–0.082). No converged profile is free of an incentive to deviate.

4.2 The punishment pattern reproduces within the pair

We first run Calvano et al.'s own test. Within co-trained pairs, a forced one-period undercut produces exactly the published pattern: the rival's price falls (mean drop 0.430 from a pre-shock level of 1.785), and in 16 of 18 pairs both prices return *exactly* to the pre-deviation level, within about five periods (Figure 3, top-left). The impulse-response evidence in Calvano et al. is not an artifact of their code or ours; it is a robust property of co-trained Q-learning pairs in this environment. The question is what it is evidence *of*.

4.3 Auditing the mechanism

A punishment scheme, in the sense of Section 2, has a signature: it responds to departures from a prescribed path, it tracks the opponent it is meant to discipline, and it makes deviating unprofitable. This section tests the converged policies for each element of that signature, in increasing order of severity: the response carries no information about *what* it is reacting to (i); it is insensitive to *whom* it is playing against (ii); the off-path policy has no best-response structure to carry a threat (iii); the celebrated return dynamic is over-determined rather than selective (iv), the pivotal test, because with a single global attractor the exhibit would follow any perturbation; and none of this is a representational limit, because hand-built punishment rules fit the same matrix exactly (v).

(i) The response carries no directional information. This test reproduces, in our population, the finding of Epivent and Lambin [2024], who show in the same environment that price hikes are followed by the same price wars as price cuts. A trigger rule may legitimately react to any departure from the prescribed path, upward included, so a response to a cooperative shock does not by itself refute path enforcement. The information is in the *form* of the response. When the forced shock goes *up* (the shocked agent raises its price to pˆM), the rival responds with the same qualitative pattern as after an undercut: a drop below baseline (to ≈ 1.62 against a base of 1.78) followed by the same return (Figure 3, bottom-left). Nothing in the response encodes the direction, the payoff relevance, or the identity of the perturbation. A response that treats a move toward the cartel price exactly like a defection is consistent with basin relaxation; it is not the signature of a rule whose function is to deter defection. Epivent and Lambin note that such a rule is admissible in equilibrium but has no counterpart in documented human collusion, and that on a grid shifted below Nash the same agents punish upward moves toward Nash; what the remaining audits add is that it is also not what the learned policy carries.

(ii) There is no working model of the opponent. If the trained agents carried a contingent retaliation rule, an opponent that actually punishes (grim trigger) should elicit different play than an opponent that never reacts (constant Nash). Designing this comparison requires care: started from a random state, a grim rule reads the initial price as a defection and reverts to Nash at once; the two opponents are then behaviorally identical, and identical play against them proves nothing. The valid design initializes the joint state at the cooperative price, so the punisher starts cooperative. The result is sharper than mere blindness: **22 of 24 trained agents**

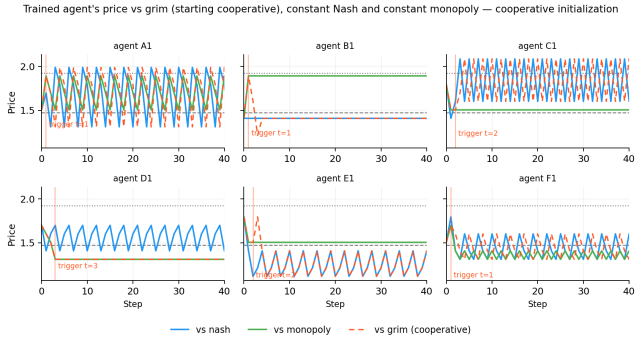


Figure 4: Trained agent's price path against a grim-trigger punisher started cooperative (dashed red), a constant-Nash pricer (blue) and a constant-monopoly pricer (green), from a cooperative initial state; six of the 24 agents shown, one panel per independent training pair (letters A–F), first agent of each pair. Vertical lines mark the period in which the agent's own undercut triggers grim's permanent reversion (22 of 24 agents, t ≤ 4).

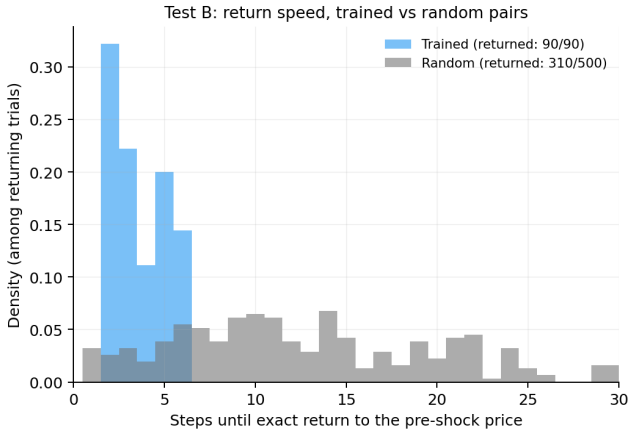


Figure 5: Null model, return speed: steps until exact return to the pre-shock price, among returning trials. Trained pairs (blue) return in 3.62 steps on average and in 90 of 90 trials; random Q-table pairs (gray) take 13.87 steps and return in 310 of 500 (62.0%). Fisher p = 1.8×10^{-17} on the rates, Mann–Whitney p = 5.8×10^{-32} on the speed; the basin shares (99.9% vs 79.2%) are reported in the text.

undercut the grim punisher within four periods, triggering the permanent reversion its rule announces, and never re-cooperate. The agents destroy a cooperative arrangement that an opponent stands ready to enforce (Figure 4). In the broader nine-strategy battery (complete over all 24 unique seeds), mean CI against fixed opponents falls to 0.146, against 0.62–0.70 for the same Q-tables with their training partners; and within every seed the CI against nash, grim, wsls and imitation is identical, expected once the punisher is triggered and symptomatic all the same.

(iii) There is no best-response logic off-path. The learned policy is a 225-entry lookup table we can inspect directly. Slicing each agent's policy at its on-path price yields its *reaction function*: price at t as a function of the rival's price at t−1. Against the clean, monotone static best-response curve, the 18 learned reaction functions are noise: mean Spearman ρ = **0.016** (range [−0.47, 0.52]). The full policy heatmaps are salt-and-pepper over the entire state space; the only economically interpretable point is the on-path attractor itself.

(iv) The return is over-determined, not selective. Here is the pivotal step. Under this state representation, the state *is* the previous joint action; two fixed greedy policies therefore define a deterministic map on 225 states, and by the pigeonhole principle every orbit is eventually periodic. Reaching *some* cycle is guaranteed by the object class; the question is whether a rule selects the destination or the map leaves nowhere else to go. Enumerating the full state space settles it: **17 of 18 trained pairs have exactly one attractor whose basin is all 225 states** (mean dominant-basin share 99.9%, mean 1.06 attractors per pair), and the exact-return rate after a forced deviation is **100% (90/90 trials)**. This is the structural reason for (i): with one globally attracting cycle, the orbit after a defection and the orbit after a cooperative shock cannot differ. A random-policy null locates the behavior: random Q-tables drawn on the same value range fragment the space (3.26 attractors, dominant basin 79.2%, Mann–Whitney p = 6.3×10^{-8}) and return only **62.0%** of the time (310/500, Fisher p = 1.8×10^{-17}), so the global basin is not a property of the state space alone. What training adds is **speed and placement**: 3.62 steps to return against 13.87 (p = 5.8×10^{-32}), and 96.7% of returns land at CI > 0.3 against 62.6% for random pairs (Figure 5). Speed and placement without selectivity are calibration, not deterrence: the trained pair returns to its path from anywhere and for any reason, which is what a basin does and not what a trigger rule does.

(v) The absence is not architectural. One might object that a 225-state lookup table cannot express punishment, making its absence trivial. We check this by construction. We hand-build two policies in the identical 225 × 15 matrix, with the same state indexing as the trained agents: a reactive rule (play Nash whenever the rival's last price was below a threshold, otherwise cooperate, so punishment lasts as long as the deviation and is then forgiven) and a grim rule (the same trigger, but once the agent's own last price is Nash it stays at Nash, so its own past price serves as a phase flag and the punishment is permanent). Simulated in the same environment against a scripted rival that cooperates, undercuts to the grid minimum for five periods and returns, from a cooperative initial state, the reactive policy punishes for exactly the five periods of the deviation and returns with the rival, and the grim policy punishes and never returns (Figure 6). Their policy heatmaps show the threshold structure as clean bands, in direct contrast with the salt-and-pepper learned policies. The representational capacity is there; the learning process did not use it.

4.4 The dyad is the unit: nothing survives outside the pair

If what training produces is a pair-specific basin rather than individual strategies, coordination should not survive rematching, just as, in partner-to-stranger comparisons in the laboratory, behavior

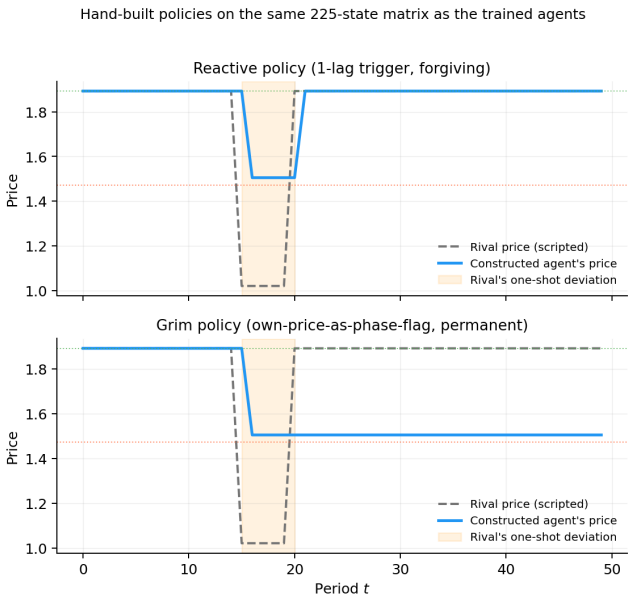


Figure 6: Capacity control: hand-built policies in the same 225 × 15 matrix as the trained agents, simulated in the same environment against a scripted rival that undercuts for five periods (shaded) from a cooperative joint state. Top: the reactive rule punishes for the duration of the deviation and forgives. Bottom: the grim rule, using its own past price as a phase flag, punishes permanently. Dotted lines mark the Nash and cooperative prices.

that is a personal norm travels and behavior that is a pair-specific convention does not. Calvano et al. [2020, Section VI.B] ran this rematch with learning on and saw profit gains fall from 85% to about 20% before re-learning partially restored them; with learning off, the collapse is the whole story. Agents trained under identical protocols but in different runs, when cross-matched and evaluated greedily, collapse from within-pair **CI = 0.674 to CI = 0.126**, an 81% loss (12 cross-matchings, all unique after the seed audit); in an extended combinatorial evaluation (40 matchings), mean CI is 0.109 and **30% of matchings price below Nash** (Figure 7). At scale the collapse is the same: 1,024 pairs trained under the same protocol on Calvano's grid give, with the long-run price averaged over all 225 initial states, CI = 0.669 within pairs and 0.134 across 2,000 random matchings, a loss of 80%; and matchings between pairs that had converged to the *same* price collapse just as far (0.138), so what fails to transfer is not the price level but the off-path structure that would return to it. The cross-paired impulse response makes the point visually: there is no plateau to defend, prices hover noisily around p^{NE}, and the forced deviation is invisible against the ambient variation (Figure 3, right column). In machine-learning terms, this is an out-of-distribution failure: each policy was fit to the distribution of states its particular partner generates, and a new partner, even one trained under an identical protocol, moves the test distribution. The pair has overfit to itself. Collusion in this

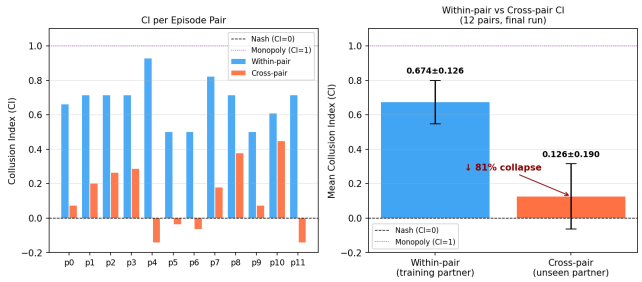


Figure 7: Within-pair vs cross-pair collusion index, per matching: CI collapses from 0.674 to 0.126 (−81%) when independently trained agents are cross-matched (12 matchings).

environment is not a strategy an agent *has*; it is a convention a pair *keeps*.

4.5 What is deterred, and what is not

Deterrence-in-effect holds for the deviation Calvano et al. test. We first run their own profitability calculation on our pairs: given what the rival actually does after the forced one-period undercut, does the deviation pay? It does not, in **18 of 18 pairs**: the one-period gain (mean +0.040) is outweighed by the discounted losses of the post-deviation phase (mean −0.240), for a mean net of **−0.200** ($\delta = 0.95$). On this point we replicate Calvano et al. exactly.

But the deterred deviation is not the relevant one. The positive CE-gap of Section 4.1 can be made concrete. Fixing each partner's actual greedy policy and solving the induced 225-state MDP exactly ($\delta = 0.95$), the optimal best response, against a partner who *reacts*, improves on converged discounted profit by **14.8% on average** (median 12.8%, maximum 48.0%; 10 of the 38 agents are at an exact best response, and excluding the duplicated pair the figures are 15.6%, 13.4% and 8 of 36) (Figure 8). This is a demanding test (no learning agent computes exact DP), and we use it not to demand optimality but to size the wedge: the profile leaves, on average, a seventh of the achievable payoff on the table against the very partner it converged with. The juxtaposition resolves the apparent tension: what the converged dyad deters is the *naive* deviation, a single-period undercut followed by a return to the old policy, which is precisely the deviation no optimizer would choose; the *optimal* deviation policy, which anticipates the partner's reaction, profits by 14.8%. Neither agent ever samples the deviations that would profit it. The supracompetitive prices are not *enforced*; they are merely *undisturbed*: collusion survives because deviation is never tried, not because it would not pay. This also locates our measure relative to the *Q-loss* of Calvano et al. [2020, Section III.D], who evaluate each limit strategy against the rival's and report about half of their sessions in on-path Nash equilibrium, with a mean on-path Q-loss of 0.2%. As we read their procedure, the Q-loss is the forfeited payoff of a one-step improvement under the agent's own continuation, checked at on-path states; our wedge is the value of the full optimal policy from the converged state, which may leave the path and keep deviating. The two are consistent: no profitable one-step deviation on path, a profitable plan off it, which is the sense in which the path is undisturbed rather than enforced. In the

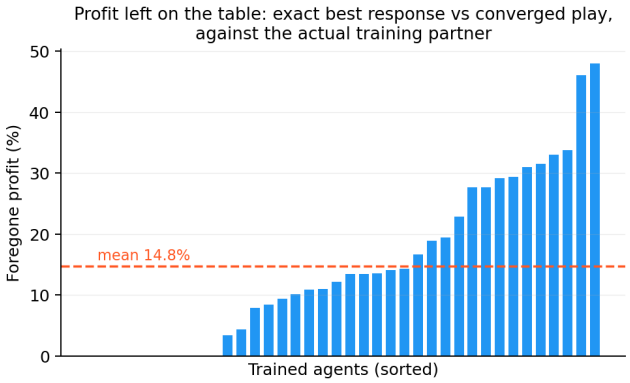


Figure 8: Exact best-response gain over converged play, per agent (38 agents): mean 14.8%, median 12.8%, maximum 48.0%; 10 of 38 agents are at an exact best response.

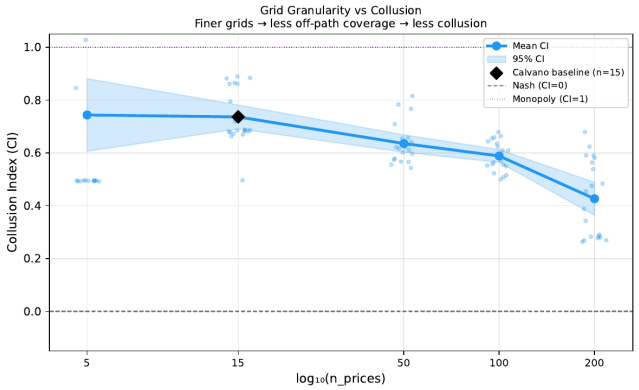


Figure 9: Grid granularity: mean CI (20 seeds per level, 2M steps) declines monotonically as the price grid is refined from 5 to 200 points; the black diamond marks Calvano's 15-point baseline.

vocabulary of Calvano IJIO and of the survey by Tang [2024], the deterrence is *incomplete*: genuine with respect to the one-period deviation they test, and spurious with respect to the optimal one, in Calvano IJIO's own sense of prices sustained by not recognizing the gain from deviating. We therefore do not claim that retaliation plays no stabilizing role against naive deviations; we claim that the role it plays does not extend to the deviation an optimizer would choose, which is the one that decides whether the path is enforced.

4.6 Why: the visitation funnel

One mechanism explains both halves of the results (why the audit finds no scheme, and why nothing survives outside the pair), and it is visible in training. As ε decays, state–action visitation concentrates on a narrow neighborhood of the joint trajectory; off-path entries of the Q-tables retain their initialization values or reflect a handful of samples. The policy outside the basin is, in the operative sense, untrained, which is why it behaves like the random-map null model wherever the basin's local calibration runs out.

The funnel makes a comparative-statics prediction: enlarging the state space, by refining the price grid, spreads the same exploration budget over more off-path states and should weaken coordination. The prediction holds. Holding the training budget at 2M steps and varying the grid from 5 to 200 points (20 seeds per level), mean CI declines monotonically from 0.74 (5 points) and 0.74 at Calvano's 15, to 0.64 (50), 0.59 (100) and 0.43 (200 points), consistent with the granularity sensitivity in Klein [2021], and prices remain supracompetitive even at the finest grid (Figure 9). The basin account thus predicts evidence beyond the audit it was built to explain. And the same exploration decay that stabilizes the collusive path removes the experience that would test it: the funnel builds the basin and then seals it.

5 Discussion

What the evidence supports. The converged objects in this environment are not individual pricing strategies sustained by contingent retaliation. They are two halves of a jointly carved attractor: fast and stable at home, structureless everywhere else. None of the signatures of a punishment scheme we test for is present, while the capacity control confirms the architecture could have expressed them. The natural theoretical name for this object is a *self-confirming equilibrium* [Fudenberg and Levine 1993]: each agent's beliefs are correct on the realized path and never tested off it, so play is optimal against beliefs that no experience contradicts, without being optimal against the opponent's actual strategy. We offer this as an interpretation, not a theorem. The dynamics that produce such an object have a candidate description in Banchio and Mantegazza [2023], in a stateless setting: the value estimates of co-learning agents are correlated through the joint action, and the correlation, not any threat, holds the profile together; the self-confirming equilibrium is what that coupling leaves behind once exploration stops. Our capacity control speaks to their conjecture that memory, by enabling reward–punishment strategies, would substantially increase collusion: with one period of memory the capacity is present and unused. We also follow den Boer et al. [2026] in preferring the language of optimality to that of threats: algorithms are neither tempted nor deterred, and asking whether a converged policy is a best response to its partner, as Section 4.5 does, is the question that remains well posed once that vocabulary is set aside. Put in Lucas-critique terms: the learned rule is calibrated to one particular environment, and here the partner *is* the environment; change the partner and the rule loses its object.

Reconciling with Calvano et al. [2023]. Calvano IJIO verify approximate incentive compatibility on their converged profiles; our exact-DP audit finds mean best-response gains of 14.8%, even though the single-deviation profitability test of the 2020 paper *replicates* on our pairs (Section 4.5). To rule out the training regime as the explanation, we retrained five pairs to Calvano's own stopping rule (no episode resets, single horizon, policy stable before termination) and reran the entire audit chain. Every result survives: the impulse-response pattern reproduces in 5 of 5 pairs, the cooperative grim punisher is undercut by 4 of 5 agents (the fifth never triggers it), the one-period deviation remains unprofitable in 5 of 5, and the exact best response still gains **10.4% on average**. Training to convergence *narrows* the incentive wedge, from 14.8% to 10.4%,

but does not close it. The remaining divergence from the "genuine collusion" classification lies primarily in the deviation class checked (one-period undercut versus exact dynamic best response), not in the training regime.

Implications for the literature and for policy. The impulse-response figure has served as the literature's strongest exhibit for treating learned supracompetitive pricing as collusion in the legal-economic sense. Our results do not overturn the figure; they reinterpret it. A drop-and-return after any perturbation is what a policy pair with a single global attractor does; its evidentiary value for *deterrence* is small without the controls run here. Studies inferring mechanisms from on-path behavior alone risk attributing to strategy what is, mechanically, basin geometry. The consumer-welfare concern, however, is not dissolved: prices are robustly supracompetitive (CI $\approx$ 0.62–0.70 within pairs), and that harm is unchanged by anything in our audit. What changes is the remedial reading. The "strategy" does not transfer to any opponent not present during training, including opponents strictly more cooperative than the training partner; and the absence of a contingent scheme cuts *against* mapping the dyad onto agreement-like legal categories (a meeting of minds enforced by threats). Our evidence licenses only a frozen-policy claim: disturbing the pairing (entry, rotation, retraining) destroys coordination among the converged policies we study. Calvano et al. [2020, Section VI.B] show that continued learning does re-carve a basin after recomposition, over tens of thousands of further periods; what our data add is that nothing is carried into that process by the policies themselves. The absence of a contingent scheme does not shrink the enforcement problem; it relocates it, from detecting threats that are not there to disturbing pairings that re-form only at the cost of re-learning at live prices, a cost the frozen test shows starts from zero coordination.

Implications for machine-learning pricing at large. Pricing decisions are increasingly delegated to learning systems, and the field evidence shows the stakes are not hypothetical: algorithm adoption measurably changes price competition [Assad et al. 2024; Brown and MacKay 2023]. For that setting our results carry a warning and a toolkit. The warning is methodological: any claim that deployed pricing agents have learned to coordinate, or to punish, inherits the evaluation problem documented here, because market data record only the on-path behavior of one particular grouping of algorithms, and Section 4 shows that on-path behavior alone cannot identify a mechanism. The same caution applies inside firms and platforms: a pricing system that performs well against the agents it was trained or tuned with has demonstrated a property of the *ensemble*, not of the policy; our cross-pairing result is a measured example of how large that gap can be. The toolkit is the audit itself: every protocol used here (stranger re-matching, directional shocks, opponents with explicit rules, random-policy null models, exploitability accounting) is defined on frozen policies and portable to any multi-agent pricing stack, including deep RL systems. Together they form a pre-deployment evaluation standard for a question that on-path monitoring cannot answer: whether apparent coordination is a strategy the agent carries, or a property of the ensemble that trained together.

Limitations. (1) *Population*: the core population is 17 unique co-trained pairs (one run of 18 was excluded as a bit-identical duplicate after a seed audit; the corrected runner later extended the population to 24 unique pairs, used where stated); the extended combinatorial cross-evaluation covers 40 of the 153 possible matchings (truncated in enumeration order, not sampled). (2) *Statistics*: the separation claims in Section 4.3(iv) are reported with pair-level aggregation; the trained arm is at a ceiling (18 of 18 pairs at rate 1.000), so the pair-level variance is zero on one side, and the interval on the rate difference, [+0.318, +0.442], should be read as a floor on the separation rather than a precise estimate. (3) *Training regime*: the main population trains a fixed 2M steps with episode resets rather than to Calvano's stability criterion; the five-pair rerun under the faithful regime replicates the entire audit chain, with the wedge narrowing to 10.4%; larger regime-faithful populations remain desirable. (4) *Scope*: our claims are about tabular Q-learning in the Calvano environment, at its baseline parameterization (one demand specification, one 15-point grid, $\delta = 0.95$), and they are bounded by it. The protocol-level audits (up-shock, opponent battery, cross-pairing) are portable to function-approximation agents; the eventual-periodicity argument behind the null model is not (it is specific to finite deterministic policy maps), and whether deep agents learn genuinely contingent responses is an open question our protocol is designed to answer, not presuppose. Within that scope, none of the signatures of deterrence we test for is present, and what is present is basin geometry.

6 Conclusion

We replicated the central exhibit of the algorithmic collusion literature and asked the question the exhibit was taken to answer: is there a punishment mechanism? None of five independent audits finds one. The response to deviation carries no directional information; faced with a punisher whose cooperation was there to be kept, the agents defect and absorb the punishment; their reaction functions are uncorrelated with best-response logic; their celebrated return-to-baseline is over-determined, since 17 of 18 pairs have a single attractor covering the whole state space; and none of this is explained by a representational limit. What training produces instead is a dyad-specific basin of attraction: re-match frozen agents from independent runs and coordination collapses by 81%, from CI = 0.674 to 0.126. And the path the basin holds is not incentive-compatible against the optimal deviation, which, computed against each partner's frozen strategy, profits by 14.8% on average. The scheme deters only the deviation no optimizer would choose. Collusion here is not a strategy either agent has; it is a convention the pair keeps. This reading unifies what the literature had measured separately: the pair-specificity Calvano et al. observed on rematching, the direction-blind response Epivent and Lambin documented, the gap between impulse patterns and equilibrium den Boer et al. established, and the punishment-free coupling Banchio and Mantegazza modeled.

The methodological point outlasts the specific exhibit. As pricing migrates to learning systems, behavioral patterns produced by black-box agents will increasingly reach regulators and courts as evidence of algorithmic coordination, and such patterns invite mechanistic over-reading: the drop-and-return *looks* like retaliation, and controls are what tell the difference. Mechanism claims about learned behavior should be held to the standard applied here: survive an explicit null, survive removal from the training context,

survive an incentive accounting. What the standard buys is precision about the policy problem. The supracompetitive prices, and the harm they imply, are robust; what is absent is the deterrence behind them, and harm and mechanism call for different instruments: an outcome can be monitored, but only an agent that carries a rule can be deterred. Prices in this environment are high not because deviation is punished, but because, once exploration has decayed, deviation is never tried, and the beliefs that would motivate it are never corrected.

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